

EXAMPLE 7**Anscombe's Quartet**

Francis Anscombe, a British statistician, developed datasets that illustrate why just looking at summary statistics (i.e., mean, standard deviation, and correlation) does not fully describe the data. He created four datasets (designated I, II, III, and IV), each with two variables, X and Y , such that:

- The X s in each dataset have the same mean and standard deviation, 9.00 and 3.32, respectively.
- The Y s in each dataset have the same mean and standard deviation, 7.50 and 2.03, respectively.
- The X s and Y s in each dataset have the same correlation of 0.82.

Exhibit 27: Summary Statistics

Observation	I		II		III		IV	
	X	Y	X	Y	X	Y	X	Y
1	10	8.04	10	9.14	10	7.46	8	6.6
2	8	6.95	8	8.14	8	6.77	8	5.8
3	13	7.58	13	8.74	13	12.74	8	7.7
4	9	8.81	9	8.77	9	7.11	8	8.8
5	11	8.33	11	9.26	11	7.81	8	8.5
6	14	9.96	14	8.1	14	8.84	8	7
7	6	7.24	6	6.13	6	6.08	8	5.3
8	4	4.26	4	3.1	4	5.39	19	13
9	12	10.8	12	9.13	12	8.15	8	5.6
10	7	4.82	7	7.26	7	6.42	8	7.9
11	5	5.68	5	4.74	5	5.73	8	6.9
N	11	11	11	11	11	11	11	11
Mean	9.00	7.50	9.00	7.50	9.00	7.50	9.00	7.50
Standard deviation	3.32	2.03	3.32	2.03	3.32	2.03	3.32	2.03
Correlation	0.82		0.82		0.82		0.82	

While the X variable has the same values for I, II, and III in the quartet of datasets, the Y variables are quite different, creating different relationships. The four datasets are:

- I An approximate linear relationship between X and Y .
- II A curvilinear relationship between X and Y .
- III A linear relationship except for one outlier.
- IV A constant X with the exception of one outlier.

Depicting the quartet visually,